

RENXI WANG

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EDUCATION

Mohamed bin Zayed University of Artificial Intelligence (MBZUAI), UAE Aug. 2025 - Present
PhD in Natural Language Processing Supervisor: Timothy Baldwin, Haonan Li, Fajri Koto
Northeastern University, China Sept. 2019 - Jun. 2023
B. Eng. in Computer Science and Technology CGPA: 90.6/100

RESEARCH INTERESTS

My research focuses on agentic reinforcement learning for large language models, with particular interest in long-horizon tasks, context folding for extended trajectories, tool-use training, and the framework infrastructure that makes scalable agent RL practical.

PAPERS & PUBLICATIONS

SimuScene: Training and Benchmarking Code Generation to Simulate Physical Scenarios [Paper] Under Review
Yanan Wang*, [Renxi Wang*](#), Yongxin Wang*, Xuezhi Liang, Fajri Koto, Timothy Baldwin, Xiaodan Liang, Haonan Li,

- Proposed SimuScene, a benchmark that evaluates LLMs capability to simulate physics
- Proposed an RL pipeline that trains a language model from video signals
- Provided benchmark, SFT, and RL datasets

ToolGen: Unified Tool Retrieval and Calling via Generation. [Paper][Code] ICLR 2025
[Renxi Wang](#), Xudong Han, Lei Ji, Shu Wang, Timothy Baldwin, Haonan Li

- Converted and represented tools as virtual tokens.
- Designed a three-stage training strategy to train a unified tool agent.
- Achieved superior performance in tool retrieval and task completion.

NAT: Enhancing Agent Tuning with Negative Samples. [Paper][Code] NAACL 2025
[Renxi Wang](#), Haonan Li, Xudong Han, Yixuan Zhang, Timothy Baldwin

- Designed tools for math reasoning task and question answering task.
- Collected interaction trajectories between agents and environments.
- Fine-tuned LLMs on collected trajectories with negative-aware training.

Demystifying Instruction Mixing for Fine-tuning Large Language Models. [Paper][Code] ACL SRW, 2024
[Renxi Wang](#), Haonan Li, Minghao Wu, Yuxia Wang, Xudong Han, Chiyu Zhang, Timothy Baldwin

- Selected different combinations of instruction to fine-tune large language models
- Analyzed how instruction datasets improve LLMs' performance.
- Simply mixing different instructions may have a negative transfer on general chat and instruction-following ability.

DialogueGLP: Global-Local Modeling with Prompt-Based Knowledge Enhancement for Emotion Inference in Conversation. [Paper][Code] EACL Findings, 2023
[Renxi Wang](#), Shi Feng

- Utilized prompt learning to generate knowledge for emotion detection.
- Designed templates for InstructGPT to generate knowledge for dialogue systems.
- Proposed new modeling method to predict emotion in conversations.

Concise Reasoning in the Lens of Lagrangian Optimization [Paper] Under Review
Chengqian Gao, Haonan Li, Taylor W. Killian, Jianshu She, *Renxi Wang*, Liqun Ma, Zhoujun Cheng, Shibo Hao, Zhiqiang Xu

A Minimalist Proof Language for Neural Theorem Proving over Isabelle/HOL [Paper][Code] OOPSLA 2026
Qiyuan Xu, *Renxi Wang*, Peixin Wang, Haonan Li, Conrad Watt

Explore the Reasoning Capability of LLMs in the Chess Testbed.[Paper] NAACL 2025
Shu Wang, Lei Ji, *Renxi Wang*, Wenxiao Zhao, Haokun Liu, Yifan Hou, Ying Nian Wu

Against The Achilles' Heel: A Survey on Red Teaming for Generative Models. [Paper][Code] JAIR
Lizhi Lin, Honglin Mu, Zenan Zhai, Minghan Wang, Yuxia Wang, *Renxi Wang*, Junjie Gao, Yixuan Zhang, Wanxiang Che, Timothy Baldwin, Xudong Han, Haonan Li

SERVICES

I have served as a reviewer for **Coling**, **ACL**, and **EMNLP**

PROJECTS & EXPERIENCES

AgentFly: Extensible and Scalable Agentic Reinforcement Learning [Project][Report] Feb. 2025 - Present
A holistic agent reinforcement learning framework, which **I lead**.

- Meticulously designed interfaces and resource layers
- Multi-task support (WebShop, SWE-Bench, Search, ScienceWorld, etc)
- Scalable to one thousand environments, hundreds of billions size of LLMs
- Paired open-source frameworks:
 - ChatBricks: LLM chat template
 - enroot-py: Asynchronous client for enroot containers

Microsoft Research Asia (MSRA) Jun. 2024 - Sept. 2024
Research Internship at System Research Group, Mentored by Lei Ji

- Trained a command agent to manipulate operating systems
- Enhanced the command agent with Reinforcement Learning
- Finished the internship with the Stars of Tomorrow Award

SKILLS

Languages: Chinese, English

Computer Skills: Python, Pytorch, Transformers, Git, DeepSpeed, verl, vLLM